

Convolution Code Snippet

```
for(int n = 0; n < 1; n+=1){
  for(int co = 0; co < 64; co+=1){
    for(int h = 0; h < 32; h+=1){
      for(int w = 0; w < 32; w+=1){
        for(int ci = 0; ci < 3; ci+=1){
          for(int kh = 0; kh < 7; kh+=1){
            for(int kw = 0; kw < 7; kw+=1){
              // convolution statement;
              ... = input[n][ci][h][w];
              .....
              output[n][co][h][w] = ...
            }
          }
        }
      }
    }
  }
}
```

Feature Extraction

Structure Pattern

Depthwise Conv

Convolution-like

Matmul-like

.....

Access Pattern

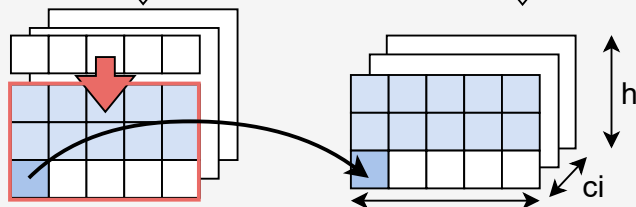
FIFO channels

input & output

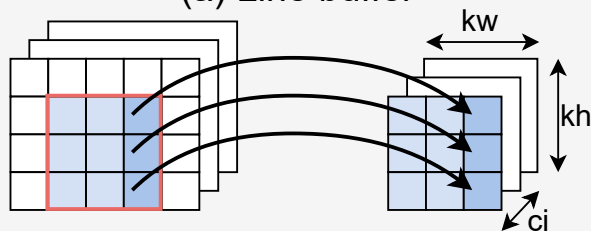
Reduction Loops

h, w, kh, kw

Buffer Generation



(a) Line buffer



(b) Window buffer

```
for (int n = 0; n < 1; n += 1) {
  for (int h = 0; h < 38; h += 1) {
    for (int w = 0; w < 38; w += 1) {
      for (int ci = 0; ci < 3; ci += 1) {
        lb[n][ci][0][w] = lb[n][ci][1][w];
        lb[n][ci][1][w] = lb[n][ci][2][w];
        .....
        lb[n][ci][6][w] = input[n][ci][h][w];
      } // Line buffer Operations
      if ((h - 6) >= 0) {
        for (int ci = 0; ci < 3; ci += 1) {
          for (int kh = 0; kh < 7; kh += 1) {
            wb[n][ci][kh][0] = wb[n][ci][kh][1];
            wb[n][ci][kh][1] = wb[n][ci][kh][2];
            .....
            wb[n][ci][kh][6] = lb[n][ci][kh][w];
          }
        } // Window buffer Operations
        if ((w - 6) >= 0) {
          float temp[64];
          for (int co = 0; co < 64; co += 1) {
            temp[co] = (float)0.000000;
            for (int ci = 0; ci < 3; ci += 1) {
              for (int kh = 0; kh < 7; kh += 1) {
                for (int kw = 0; kw < 7; kw += 1) {
                  float v1 = wb[n][ci][kh][kw];
                  float v2 = weight[co][ci][kw][kw];
                  .....
                  temp[co] = v5;
                }
              }
            }
            float v1024 = v1014[v1015];
            out[n][co][(h-6)][(w-6)] = temp[co];
          }
        }
      }
    }
  }
}
```

LineBuffer

WindowBuffer

Convolution

(c) Transformed convolution code